ROOT CAUSE UNLOCKED · CONDITION KIT

The Reflux Root Cause *Kit*

Which tests actually matter, when you need a scope instead, and exactly what to ask your doctor for

Dr. Adam Seay, DC

Seay Wellness / Root Health, Garner, NC · rootcauseunlocked.com · (919) 662-0520

The Reflux Root Cause Kit

Which tests actually matter, when you need a scope instead, and exactly what to ask your doctor for

> **BEFORE YOU START**

Contents

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Some reflux symptoms are not reflux. Get medical attention now if you have:

- Chest pain, pressure, or tightness, especially with sweating, nausea, shortness of breath, or pain into the jaw or arm. **Heart attacks are mistaken for heartburn every day. Call 911.**
- Vomiting blood, or stool that is black and tarry
- Food that sticks or will not go down. Complete inability to swallow your own saliva is an emergency.

Book a doctor's appointment, do not self-treat, if you have any of these:

- Difficulty swallowing, or the sensation of food catching
- Unintentional weight loss
- Persistent vomiting
- Anemia, or a known low iron level
- Symptoms that began after age 60
- A family history of esophageal or stomach cancer

Those are the alarm features. Guidelines put patients aged 60 and over presenting with dyspepsia straight to upper endoscopy to exclude organic disease, and note that people at higher risk of malignancy may warrant endoscopy at a younger age (PMID 28631728). This matters more in reflux than in almost anything else I write about, because reflux has a serious downstream complication and because the symptoms are easy to normalize for years.

The honest starting point, and the five drivers

There is no blood test for reflux. Reflux is diagnosed clinically, and confirmed when needed by endoscopy or pH testing, both of which are specialist procedures (PMID 39692638).

That is not a reason to skip testing. It is a reason to test the right thing. Acid is what makes reflux *hurt*. Acid is almost never what makes reflux *happen*. If you only ever measure and suppress the acid, you are treating the last step in a chain and leaving the first four in place, which is exactly why up to half of people remain symptomatic despite adequate acid suppression, a figure that has pushed even the mainstream literature toward describing reflux as a multifactorial condition involving anti-reflux barrier dysfunction rather than a simple acid problem (PMID 42198379).

The five drivers, and what tests each one

Everything in this guide is organized around the same five drivers used in *Unlock Your Reflux*, because the point of testing is to find out which of them is yours.

Driver	What it means	How you find it
1. The barrier	The valve at the top of the stomach is mechanically compromised: sphincter, diaphragm, or hiatal hernia	Endoscopy, symptom pattern, breathing and postural assessment. Not a blood test
2. The signal	The nervous system that runs the valve and reports the pain is dysregulated: vagal tone, stress, sleep, hypervigilance	History, sleep assessment, HRV. Not a blood test
3. The terrain	The microbial environment of the stomach and small intestine is disturbed: H. pylori, overgrowth, dysbiosis	H. pylori breath or stool test, breath testing, stool analysis
4. The load	What is being asked of the system: meal size, meal timing, weight, food reactivity, histamine	Food and symptom diary, structured elimination and reintroduction
5. The capacity	Whether the stomach is doing its job at all: acid, enzymes, emptying	Autoimmune gastritis panel, B12 and iron, gastrin and pepsinogens

Read that table as the map for the rest of this guide. Two of those five drivers, the barrier and the signal, are the ones most often missed, and neither of them has a blood test. That does not make them less real. It makes them the ones you have to look for deliberately.

The blood work in this guide does three jobs inside that map:

1. **Find a treatable cause underneath the reflux.** H. pylori is the big one, and it sits in the terrain.
2. **Find the conditions that imitate reflux** and are treated completely differently. Eosinophilic esophagitis and celiac disease belong here.
3. **Find the consequences of the reflux, or of its treatment.** Iron, B12, and magnesium all belong here, and this is the part most often skipped.

The complication you should understand before anything else

Long-standing reflux can change the lining of the lower esophagus into a different type of tissue, called Barrett's esophagus. It is a precancerous change, it results from chronic reflux, and it raises the risk of esophageal adenocarcinoma, a cancer whose incidence has been rising in Western populations (PMID 41230608).

I am not telling you this to frighten you into a scope. Most people with reflux never develop it. I am telling you because it is the reason that "I have had heartburn for fifteen years and I just live with it" is not a neutral choice, and it is the reason the alarm features above are worth taking seriously rather than managing around with antacids.

Driver 1: the barrier, which is mechanical and has no blood test

This is the part of reflux that no lab will ever show you, and for many people it is the whole answer.

What the anti-reflux barrier actually is. Not one valve, but three components working together: the lower esophageal sphincter, the crural diaphragm wrapped around it, and the flap valve formed by the angle at which the esophagus enters the stomach (PMID 40412989).

The diaphragm is half of your sphincter. The crural diaphragm acts as an external sphincter: its contractions raise lower esophageal sphincter pressure with every inspiration and during any rise in abdominal pressure. A sliding hiatus hernia pulls the sphincter up and out of the diaphragm's grip, and it is associated with reflux symptoms, increased acid exposure, and the severity of esophagitis (PMID 15316404). Most reflux episodes are not caused by a permanently weak sphincter at all, but by transient sphincter relaxations, and losing crural support through a hernia significantly compromises the whole barrier (PMID 11003803).

Why this matters more than any supplement in this kit. If your diaphragm is the external sphincter, then how you breathe and how you hold your trunk are not adjunct wellness advice. They are direct inputs to the valve. That is the physiological basis for the breathing work in the supplement guide and for the postural chapter of the ebook, and it is why a chiropractic and structural assessment belongs in a reflux workup rather than being an unrelated appointment.

How the barrier gets assessed.

- **Endoscopy** grades hiatal disruption directly, and there is now a formal endoscopic grading of hiatus integrity that correlates with how well the barrier is holding on prolonged pH recordings (PMID 40412989). If you are having a scope anyway, this is worth asking about by name.
- **Your symptom pattern is informative.** Symptoms tied to bending, lifting, tight waistbands, coughing, straining on the toilet, or a full abdomen point toward pressure and barrier mechanics rather than acid quantity.
- **A breathing and postural assessment.** Whether you breathe into your ribs and belly or into your neck and shoulders, whether your thoracic spine extends, and what happens to your abdomen under load.

One honest note on the emerging piece. A randomized controlled trial of structured inspiratory muscle training in reflux, aimed squarely at the diaphragmatic half of the barrier, has been designed and registered, and it is **published as a protocol with results not yet reported** (PMID 41874149). The rationale is sound and the outcome data are pending. I would rather tell you that than imply the trial already exists.

Driver 2: the signal, where the nervous system sets the volume

The same event can be a minor sensation in one person and disabling pain in another. That difference lives in the nervous system, and it is a genuine physiological finding rather than a way of saying the pain is imaginary.

Stress changes the experience of reflux. In 42 patients undergoing 24-hour pH monitoring, a psychological stressor produced increased subjective symptom ratings, and the authors conclude that the perception of symptoms **in the absence of increased reflux** during stress may explain low response rates to traditional treatment (PMID 16310024). In other words, the acid did not necessarily rise; the volume knob did.

Hypervigilance is measurable and it predicts severity. Esophageal hypervigilance is present across reflux presentations regardless of acid burden or symptom index, and it significantly predicts symptom severity even after controlling for symptom-specific anxiety (PMID 33432708). This is why two people with identical pH studies can have completely different lives.

And the honest counterweight, because it exists. Not every experiment agrees. When acute stress was induced with an IQ test during esophageal acid perfusion in 15 healthy volunteers and 10 reflux patients, it did **not** increase acid perception, even though blood pressure confirmed the stressor worked (PMID 19453516). Short laboratory stress is not the same as months of chronic load, and the honest summary is that chronic stress and hypervigilance have better support here than a single acute stressor does.

The intervention evidence is the encouraging part, and it is in the supplement guide: relaxation training reduced both symptom reports and total esophageal acid exposure compared with an attention-placebo control (PMID 8020690). A mind-body intervention moved an objective acid measurement. That is not a small finding.

Sleep belongs in this driver, and it runs both ways. A systematic review and meta-analysis found a bidirectional relationship between reflux and sleep problems, with reflux associated with poor sleep quality (OR 1.47), sleep disturbance (OR 1.47), and short sleep duration (OR 1.17) (PMID 38646475). Reflux wrecks sleep, and poor sleep worsens reflux, and you can enter that loop from either end.

Sleep apnea deserves its own line. Reflux and sleep apnea are linked, with proposed mechanisms including acid and pepsin irritation of the airway and vagally mediated changes in bronchial tone (PMID 40045407). If you snore, wake unrefreshed, or wake choking, ask for a sleep assessment. Treating the apnea can improve the reflux, and no supplement substitutes for that.

How the signal gets assessed. Not by blood work. By history, by a sleep assessment, by tracking symptoms against stress and sleep rather than against food alone, and by heart rate variability if you want a number to follow. HRV from a wearable is imperfect and it is a reasonable way to watch vagal tone respond to the breathing practice over weeks.

Driver 3: the terrain, and the blood work worth running

Helicobacter pylori

Why it is first. It is common, it is treatable, and it causes real disease. Roughly one in five peptic ulcers is associated with H. pylori infection, and the combination of H. pylori and NSAID use increases bleeding ulcer risk more than sixfold. The test-and-treat strategy is the mainstay of outpatient management: patients under 60 with dyspepsia and no alarm symptoms should be tested and treated if positive (PMID 36791443).

Which test. The two non-invasive options are the urea breath test and the stool antigen test. In a comparison against endoscopy, the 13C urea breath test had higher positive and negative predictive values (84.2% and 90.9%) than the stool antigen test (81.3% and 71.4%), with an accuracy of 86.7% versus 76.7%, and the two agreed closely when used to confirm successful eradication (PMID 32635179).

The rule that ruins this test. Both tests are affected by acid suppression and antibiotics. **You generally need to be off PPIs for about two weeks and off antibiotics and bismuth for about four weeks before testing**, or you will get a false negative. Ask your clinician for the exact interval and do not stop a prescription without telling them.

Serology is the weak option. A blood antibody test can stay positive after successful treatment, so it cannot distinguish current from past infection, and it is not the test to use for confirming eradication.

If you are treated, confirm it worked. Retest at least four weeks after finishing therapy. Clarithromycin resistance is rising, and first-line treatment has changed accordingly (PMID 36791443). Eradication is not assumed, it is verified.

Iron studies and a complete blood count

Two reasons, pulling in opposite directions.

Bleeding. An ulcer or an inflamed esophagus can bleed slowly enough that you never see it, and iron deficiency anemia may be the first sign. That is an alarm feature and it needs a doctor.

Absorption. Long-term acid suppression reduces the absorption of several nutrients. Reviews of long-term PPI use list iron alongside magnesium, calcium, and vitamin B12 among the micronutrient deficiencies associated with chronic use (PMID 36791443, PMID 41010960).

Ask for ferritin, serum iron, TIBC, transferrin saturation, and a CBC together.

Vitamin B12 and magnesium, the two the medication takes

B12. Stomach acid is required to liberate B12 from food protein, which is why long-term acid suppression is associated with B12 deficiency (PMID 36791443). If you have been on a PPI for more than a year and nobody has checked, this is the single most reasonable request you can make. B12 deficiency causes fatigue and neurological symptoms that get attributed to almost anything else.

Magnesium. Chronic PPI use is associated with hypomagnesemia, and it is on the standard list of long-term PPI harms (PMID 41010960, PMID 36791443). It can cause cramps, palpitations, and in severe cases arrhythmia. Serum magnesium is an imperfect measure of total body status, and it is still the test you will be able to get. Ask for it if you have been on acid suppression long-term.

The imitators, which must be excluded before anything else

These matter because they are treated completely differently, and because reflux is the default label that gets applied before anyone looks.

Eosinophilic esophagitis

An allergic inflammation of the esophagus that presents with reflux-like symptoms, difficulty swallowing, and food getting stuck. **It is diagnosed only by biopsy at endoscopy**, so no blood test rules it in or out.

Why it deserves a section: esophageal food impaction is the leading complication in people with undiagnosed eosinophilic esophagitis, and in a multicenter study of patients presenting with food impaction, biopsies were taken in only 52% of cases, yet 68.9% of those biopsied received a new diagnosis of the condition (PMID 40971831).

Read that number again. In the people who got the biopsy, most of them had it. **If you have ever had food stick in your esophagus badly enough to need help, ask specifically whether esophageal biopsies were taken.** If they were not, that is a reasonable question to raise.

Celiac disease

Reflux and upper gut symptoms are a recognized presentation. The serology must be drawn while you are still eating gluten, or it is uninterpretable. Removing gluten before testing is the most common way this diagnosis gets missed for years.

Cardiac disease

Not a lab, and the most important item on this page. Reflux and cardiac pain overlap in presentation, and reflux is the more comfortable explanation, which is exactly why it is the dangerous one. New chest pain deserves a cardiac evaluation before it is filed under heartburn, particularly with exertion, and particularly if you have cardiovascular risk factors.

Driver 5: capacity, and how much weight to put on the acid question

Stomach acid, and the honest state of that question

Functional medicine often frames reflux as a problem of **too little** stomach acid rather than too much, and the reasoning is mechanistic: inadequate acid means poor sphincter signaling, slower gastric emptying, and incomplete protein digestion.

Here is where the evidence actually sits, stated plainly so you can judge it.

The strongest published case for replacing lost stomach acid is made for **autoimmune gastritis**, a defined condition in which the parietal cells that make acid are destroyed, producing documented hypochlorhydria and eventually achlorhydria. In that population, the argument is that B12 and iron are routinely replaced while the missing acid is not, that these patients are often prescribed acid suppression for a stomach that is not producing acid, and that re-acidification could improve gastrointestinal symptoms and may theoretically reduce risks tied to hypergastrinemia and N-nitroso compound formation (PMID 38474790).

What that supports and what it does not. It supports acid replacement in someone with **demonstrated** low acid production. It is not evidence that most people with reflux have low stomach acid, and that broader claim does not have trial support behind it.

So the practical position is: test rather than assume. If low acid is genuinely the mechanism in your case, that should be demonstrable, most convincingly through evaluation for autoimmune gastritis, which includes parietal cell and intrinsic factor antibodies, gastrin, pepsinogen I and II with their ratio, plus B12 and iron. A raised gastrin with a low pepsinogen I to II ratio, in someone with B12 or iron deficiency, is a real pattern rather than an inference.

The at-home betaine HCl challenge, where you increase capsules until you feel warmth, is widely described and is not a validated test. It can be informative in practice and it should not be treated as a diagnosis, particularly when the same symptoms would be produced by an ulcer.

The esophageal and gastric microbiome

The terrain thinking that functional medicine has applied to the gut for years has now reached the esophagus, and the mainstream literature is catching up.

A review of the microbiome in esophageal disease describes the picture from the last five years of sequencing work, while being candid that antibiotic use, PPI use, dietary habits, and geography all drive variability and make standardized study difficult (PMID 42132964). A study sampling oral, gastric, and stool microbiomes in patients with reflux esophagitis and Barrett's esophagus found compositional differences between patients and controls, including an association between salivary Tannerellaceae and elevated serum IL-6 (PMID 41486021).

The most useful version of this for you is what it implies about long-term acid suppression. A narrative review from a microbial ecology perspective points out that up to 50% of people remain symptomatic despite adequate acid suppression, describes reflux as multifactorial rather than acid-centric, and examines how suppressing acid alters the microbial ecology of the upper gut (PMID 42198379).

Read that as directional rather than actionable: there is not yet a microbiome test that tells you what to do about your reflux. What it does support is the underlying premise, that acid suppression is not a neutral act and that the environment of the stomach matters beyond its pH.

Small intestinal bacterial overgrowth

A case-control study of 50 patients with GERD confirmed by manometry and 24-hour acid monitoring, against 53 symptom-matched controls, found SIBO positively correlated with GERD, and mean expiratory hydrogen significantly higher in the GERD group. The authors describe a potential connection and a new perspective on the pathogenesis (PMID 41131458).

Read that as an association in 103 people, not a causal chain. Breath testing is reasonable if bloating and distension dominate your picture. It is not where I would start in someone whose main complaint is heartburn.

The rest of the terrain

Thyroid function, fasting glucose and HbA1c, and hs-CRP are all cheap, and they occasionally reframe the picture. Body weight belongs here too, and it is covered in the diet guide, because it has better evidence behind it than any test on this page.

Functional ranges for the markers in this guide

Clinical targets used at Root Health, not validated diagnostic thresholds, narrower than the standard laboratory range on purpose.

Marker	Units	Functional target	Approximate standard range
Ferritin	ng/mL	50 to 150	25 to 300
Serum iron	ug/dL	65 to 110	40 to 150
TIBC	ug/dL	290 to 370	250 to 400
Transferrin saturation	%	20 to 40	15 to 45
Vitamin B12	pg/mL	500 to 900	200 to 1100
Folate (RBC)	ng/mL	600 to 1500	400 to 2000
Magnesium (serum)	mg/dL	2.0 to 2.5	1.5 to 2.7
Magnesium (RBC)	mg/dL	5.2 to 6.8	4.0 to 7.2
Zinc (plasma)	ug/dL	70 to 110	60 to 120
Vitamin D (25-OH)	ng/mL	40 to 80	20 to 100
hs-CRP	mg/L	under 0.5	under 1.0
Glucose, fasting	mg/dL	75 to 86	70 to 99
HbA1c	%	4.6 to 5.3	4.0 to 5.6
Albumin	g/dL	4.2 to 4.8	3.5 to 5.0
White blood cells	K/uL	5.0 to 7.5	3.5 to 10.5

Reading your results as a pattern

Pattern one: H. pylori positive. The clearest good news in this guide, because it is treatable and verifiable. Treat, then confirm eradication at least four weeks later.

Pattern two: long-term PPI with depleted nutrients. Low B12, low ferritin, low magnesium in someone who has been on acid suppression for years. Both halves need attention: replace what is low, and have a real conversation about whether the medication is still needed at the current dose. The deprescribing section of the supplement guide covers how that is done safely.

Pattern three: everything normal, symptoms persist. Common. It moves the work to the things labs do not measure: weight, meal timing, the position you sleep in, and the breathing and vagal work in Chapter 4 of the ebook. Those are covered in the diet guide, and several of them outperform anything on this lab list.

Pattern four: normal labs, mechanical story. Symptoms on bending, lifting, tight clothing, or a full stomach, worse lying flat, better standing. That is a barrier pattern, and the work is mechanical: bed elevation, meal size, breathing, posture, and a conversation about hiatal anatomy.

Pattern five: normal labs, nervous system story. Symptoms track stress and poor sleep more than they track food, and the intensity is out of proportion to what any scope shows. That is the signal driver, and hypervigilance predicts symptom severity independently of acid burden (PMID 33432708). The work is vagal, behavioral, and sleep-focused, and it is not a lesser diagnosis.

Pattern six: swallowing trouble. Stop reading and get the endoscopy. Difficulty swallowing is not a terrain problem.

The conversation with your doctor

A request list you can bring

- H. pylori testing by urea breath test or stool antigen, after the appropriate washout from acid suppression
- CBC, ferritin, serum iron, TIBC, transferrin saturation
- Vitamin B12, and RBC folate
- Magnesium, if you have been on acid suppression long-term
- Celiac serology, drawn while still eating gluten
- TSH with free T4
- 25-hydroxy vitamin D
- A discussion of upper endoscopy if you have any alarm feature, are 60 or over with new symptoms, or have had symptoms for many years

Usually through a functional medicine practice, often out of pocket:

- Autoimmune gastritis panel (parietal cell and intrinsic factor antibodies, gastrin, pepsinogen I and II with the ratio), if low stomach acid is genuinely suspected
- Breath testing for bacterial overgrowth, if bloating and distension dominate
- Comprehensive stool analysis, if the terrain picture is broader than reflux alone

Not laboratory tests at all, and for many people the highest-yield part of the workup:

- **A breathing and postural assessment.** Whether you can breathe diaphragmatically, what your thoracic spine does, and what happens to abdominal pressure under load. This assesses Driver 1 directly and no blood test substitutes for it
- **If you are having an endoscopy anyway, ask about the hiatus specifically**, since hiatal integrity can be graded endoscopically and correlates with barrier competence (PMID 40412989)
- **A sleep assessment**, formal if you snore or wake unrefreshed
- **Two weeks of tracking symptoms against stress and sleep**, not only against food

The sentence that changes the appointment

“I have been on acid suppression for X years, and I would like to know whether I still need it and whether it has affected my B12, iron, or magnesium.” That is a specific, reasonable, guideline-aligned request, and it opens the deprescribing conversation without sounding like you are refusing treatment.

What these labs cannot tell you

- **Whether you have reflux.** That is clinical, and confirmed by scope or pH study when it matters (PMID 39692638).
- **Whether you have Barrett's esophagus.** Only endoscopy answers that (PMID 41230608).
- **Whether a specific food is your trigger.** Only a structured elimination and reintroduction does, and the diet guide covers it.
- **Whether your stomach acid is low.** Not from a routine panel. That takes the specific workup described above.

When to seek clinical review

Emergency: chest pain with sweating, nausea, breathlessness, or pain into the jaw or arm. Vomiting blood or black tarry stools. Inability to swallow saliva.

Soon:

- Difficulty swallowing or food catching, ever
- Unintentional weight loss, persistent vomiting, or anemia
- New symptoms starting at age 60 or over
- Symptoms needing daily treatment for more than a few weeks
- Ten or more years of untreated symptoms, which is a Barrett’s screening conversation
- Reflux with cough, hoarseness, or asthma that is not improving

Your next step

Testing tells you what you are dealing with. The companion guides do the rest: **The Reflux Diet Guide** covers what actually has evidence behind it, including which famous trigger foods do not, and **The Reflux Supplement Guide** covers what has been studied and how deprescribing is done safely.

> ***Book a free Root Discovery Session: rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=reflux-labs***

Consultations are available to people located in North Carolina, where Dr. Seay is licensed.

If you are outside North Carolina, start a symptom and meal diary. In reflux it is unusually informative, because the timing between a meal, a position, and a symptom is the diagnosis:

> ***Free symptom tracker: rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=reflux-labs***

References

All references were verified against live PubMed records on 2026-08-08, including a publication type and corrections check for retractions and errata. Verification detail is documented in `Reflux-Kit-Citation-Ledger.md`.

1. Moayyedi P, Lacy BE, Andrews CN, et al. ACG and CAG Clinical Guideline: Management of Dyspepsia. *Am J Gastroenterol.* 2017;112(7):988-1013. PMID 28631728

2. ASGE Standards of Practice Committee, Desai M, Ruan W, et al. American Society for Gastrointestinal Endoscopy guideline on the diagnosis and management of GERD: summary and recommendations. *Gastrointest Endosc.* 2025;101(2):267-284. PMID 39692638

3. Al Hazzouri A, Attieh P, Azizi T, et al. Barrett's Esophagus. *J Gastroenterol Hepatol.* 2026;41(1):41-53. PMID 41230608

4. McConaghy JR, Decker A, Nair S. Peptic Ulcer Disease and H. pylori Infection: Common Questions and Answers. *Am Fam Physician.* 2023;107(2):165-172. PMID 36791443

5. Alzoubi H, Al-Mnayyis A, Al Rfoa I, et al. The Use of 13C-Urea Breath Test for Non-Invasive Diagnosis of Helicobacter pylori Infection in Comparison to Endoscopy and Stool Antigen Test. *Diagnostics (Basel).* 2020;10(7):448. PMID 32635179

6. Andrawes M, Andrawes W, Das A, et al. Proton Pump Inhibitors, An Evidence-Based Review of Indications, Efficacy, Harms, and Deprescribing. *Medicina (Kaunas).* 2025;61(9). PMID 41010960

7. Siegfried Y, Pokryszka J, Hruz P, et al. Wide discrepancy between best practice recommendations and real-life management of suspected eosinophilic esophagitis-associated food bolus impaction. *Dis Esophagus.* 2025;38(5). PMID 40971831

8. Taylor L, McCaddon A, Wolffenbuttel BHR. Creating a Framework for Treating Autoimmune Gastritis, The Case for Replacing Lost Acid. *Nutrients.* 2024;16(5):662. PMID 38474790

9. Hu Z, Shan B, Wang Y, et al. Association of gastroesophageal reflux disease with small intestinal bacterial overgrowth: a case-control study. *BMC Gastroenterol.* 2025;25(1):753. PMID 41131458

10. Kahrilas PJ, Nguyen NT. The Esophagogastric Junction Reconsidered. *Gastrointest Endosc Clin N Am.* 2025;35(3):541-553. PMID 40412989

11. van Herwaarden MA, Samsom M, Smout AJ. The role of hiatus hernia in gastro-oesophageal reflux disease. *Eur J Gastroenterol Hepatol.* 2004;16(9):831-835. PMID 15316404

12. Holloway RH. The anti-reflux barrier and mechanisms of gastro-oesophageal reflux. *Baillieres Best Pract Res Clin Gastroenterol.* 2000;14(5):681-699. PMID 11003803

13. Syropoulos S, Moutzouri M, Grammatopoulou E, et al. Impact of Inspiratory Muscle Training in Individuals with Gastroesophageal Reflux Disease: A Randomized Controlled Trial Protocol. *Methods Protoc.* 2026;9(2). PMID 41874149

14. Wright CE, Ebrecht M, Mitchell R, et al. The effect of psychological stress on symptom severity and perception in patients with gastro-oesophageal reflux. *J Psychosom Res.* 2005;59(6):415-424. PMID 16310024

15. Guadagnoli L, Yadlapati R, Taft T, et al. Esophageal hypervigilance is prevalent across gastroesophageal reflux disease presentations. *Neurogastroenterol Motil.* 2021;33(8):e14081. PMID 33432708

16. Hemmink GJ, Bredenoord AJ, Weusten BL, et al. Does acute psychological stress increase perception of oesophageal acid? *Neurogastroenterol Motil.* 2009;21(10):1055-e86. PMID 19453516

17. McDonald-Haile J, Bradley LA, Bailey MA, et al. Relaxation training reduces symptom reports and acid exposure in patients with gastroesophageal reflux disease. *Gastroenterology.* 1994;107(1):61-69. PMID 8020690

18. Tan X, Wang S, Wu F, et al. Bidirectional correlation between gastroesophageal reflux disease and sleep problems: a systematic review and meta-analysis. *PeerJ.* 2024;12:e17202. PMID 38646475

19. Wang J, Wang P, Lv J, et al. Exploring the silent connection: unveiling the intricate relationship between gastroesophageal reflux disease and sleep apnea syndrome. *Hum Genomics.* 2025;19(1):23. PMID 40045407

20. He P, Stein E. The Microbiome and Esophageal Disease: Where Are We Now? *Curr Gastroenterol Rep.* 2026;28(1). PMID 42132964

21. Yao CC, Tai WC, Liang CM, et al. Alternation of the oral, gastric and stool microbiome in patients with reflux esophagitis and Barrett's esophagus. *J Microbiol Immunol Infect.* 2026;59(4):463-470. PMID 41486021

22. Zanoni A, Facchin S, Mari V, et al. Rethinking Long-Term PPI Therapy in GERD: A Narrative Review from a Microbial Ecology Perspective Beyond Acid Suppression. *Pharmaceuticals (Basel).* 2026;19(5). PMID 42198379

Dr. Adam Seay, DC · Seay Wellness / Root Health, Garner, NC
rootcauseunlocked.com · (919) 662-0520

Educational content only. Not medical advice. Not a diagnosis. Not a substitute for your clinician. Chest pain is a medical emergency until proven otherwise.

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Call 911 for chest pain or pressure with sweating, nausea, breathlessness, or pain into the jaw or arm. Heartburn and heart attack are confused constantly, and the comfortable explanation is the dangerous one.

See a doctor before dieting your way through this if you have trouble swallowing, food sticking, unintentional weight loss, vomiting blood, black tarry stools, anemia, or new symptoms after age 60. Those are covered in the companion lab guide and none of them are a diet problem.

If you are pregnant, underweight, diabetic, or have a history of disordered eating, talk with a clinician before restricting food or changing meal timing.

Where food actually sits in the reflux picture

Reflux has five drivers, and they are the spine of this whole kit: **the barrier** (the valve and the diaphragm), **the signal** (vagal tone, stress, sleep), **the terrain** (the microbial environment), **the load** (what you are asking the system to handle), and **the capacity** (whether the stomach is doing its job). The lab guide maps all five.

Food is the fourth. That placement is deliberate and it is the most useful thing in this guide, because most people arrive believing food is the first, spend two years subtracting foods, and never touch the driver that was actually theirs.

Food still matters, and it matters in a specific way: **it changes the load on a barrier that is already compromised.** A large meal distends the stomach and triggers the sphincter relaxations that let reflux happen. A late meal puts that load in your esophagus while you are horizontal. Extra abdominal weight raises the pressure the valve has to hold against. Those are mechanical effects of food, and they are where the evidence is.

What the evidence does not support is the other model, the one where specific foods are chemically toxic to your esophagus and must be eliminated forever.

Most of what you have been told to avoid has never been shown to help.

That sentence will save some readers years. When researchers applied an evidence grading system to the standard lifestyle advice for reflux, reviewing the literature on smoking, alcohol, obesity, weight loss, caffeine and coffee, citrus, chocolate, spicy food, head-of-bed elevation, and late evening meals, they found **no published evidence of efficacy for dietary measures**. What survived the review was weight loss and head-of-bed elevation (PMID 16682569).

A later systematic review reached a compatible place with more nuance: weight loss and smoking cessation should be recommended to people with reflux who are obese or who smoke respectively, and for nocturnal reflux, avoiding late evening meals and elevating the head of the bed are effective. In randomized trials, late evening meals increased time with supine acid exposure compared with early meals by 5.2 percentage points, and head-of-bed elevation decreased supine acid exposure time from 21% to 15% (PMID 25956834).

So the hierarchy is not what the trigger lists imply. It looks like this:

1. **Weight, if you carry extra.** The single most evidenced dietary lever.
2. **When you eat, and how you sleep.** Timing and position beat food choice.
3. **Your own individual triggers**, discovered by testing rather than assumed from a list.
4. **The general pattern of the diet**, which is worth improving for reasons beyond reflux.

Everything below follows that order, because spending your effort in that order is what gets results.

Part one: the two things with the best evidence

Weight

If you are carrying extra weight, particularly around the middle, this is the highest-yield dietary change available to you. Abdominal weight increases pressure on the stomach and the junction between stomach and esophagus, which is mechanical rather than metaphorical. Weight loss improved both acid measurements and symptoms in the evidence review, one of only two measures that did (PMID 16682569).

Two honest notes. First, this is not a reason to attempt something drastic; gradual loss is what the evidence describes and crash approaches make reflux worse in the short term. Second, if this does not apply to you, skip it entirely rather than feeling you have missed the main event. Plenty of people with reflux are not overweight.

Timing and position

Sleep is part of this, in both directions. A systematic review and meta-analysis found a bidirectional relationship between reflux and sleep problems: reflux is associated with poor sleep quality, sleep disturbance, and short sleep duration, and the relationship runs the other way too (PMID 38646475). You can enter that loop from either end, which means fixing the evening meal and fixing the sleep are the same project rather than two.

Stop eating three hours before you lie down. This is the highest-value free change in the entire guide, and in the randomized data late meals measurably increased the time acid sat in the esophagus overnight (PMID 25956834).

Raise the head of the bed. Not extra pillows, which bend you at the waist and can make it worse. Raise the head of the bed frame itself by six to eight inches, or use a wedge under the mattress. Head-of-bed elevation cut supine acid exposure time from 21% to 15% (PMID 25956834).

Sleep on your left side. The left lateral position improved the time esophageal pH stayed above 4 in the evidence review (PMID 16682569). It is anatomy: on the left, the junction sits above the pool of stomach contents rather than in it.

Eat smaller meals. A large meal distends the stomach, and a distended stomach triggers the sphincter relaxations that let reflux happen. Smaller and more frequent beats large and occasional here.

Smoking, with the honest nuance

Smoking cessation is worth doing, and the reflux-specific evidence is narrower than you would expect. In a prospective population cohort of 29,610 people, stopping smoking was associated with improvement in **severe** reflux symptoms only among people of normal body mass index who were using antireflux medication at least weekly, with an odds ratio of 5.67 in that group, and not among overweight individuals or those with minor symptoms (PMID 24322837).

Stop smoking. Just know that if you are overweight, the weight is likely the bigger reflux lever of the two.

Part two: finding your actual triggers

The standard trigger list is coffee, chocolate, citrus, tomato, spicy food, mint, fat, and alcohol. As a general prescription it did not survive evidence review (PMID 16682569). As a list of **candidates to test on yourself**, it is perfectly reasonable, because individual reactions are real even when the population-level evidence is not.

The difference between those two uses is the whole game.

The two-week test, run properly

Days 1 to 7, baseline. Change nothing. Record every meal with a time, and every symptom with a time and a severity out of ten. The gap between the meal and the symptom is the most informative number you will collect.

Look for the pattern before you remove anything. Most people find their diary implicates two or three items, not fifteen, and often it implicates a *timing* or a *portion size* rather than a food.

Days 8 to 21, remove your top three candidates. Only three. Keep everything else stable.

Then reintroduce, one at a time. A normal portion on one day, then two days of watching. A food that reproduces symptoms across two separate trials is a real trigger for you. Everything else goes back in.

Expect a short list. If you finish this process avoiding twelve foods, something has gone wrong with the method rather than with your esophagus.

What is worth knowing about the usual suspects

Alcohol. Relaxes the lower esophageal sphincter and is one of the more commonly reported precipitants. Worth removing during any trial period so it does not confound your results.

Coffee. Frequently blamed, and the population evidence for avoidance is weak (PMID 16682569). Test it. Many people find that the amount and the timing matter more than the coffee itself, particularly coffee on an empty stomach.

Fat and portion size. Large, fatty meals slow gastric emptying and keep the stomach distended for longer. In practice I see portion size matter more often than fat content.

Mint and chocolate. Both are traditionally implicated on sphincter-relaxation grounds. Both belong in the test-it category rather than the avoid-it-forever category.

Carbonation. Frequently reported and easy to test. If fizzy drinks reliably produce symptoms within the hour, you have your answer without a study.

Part two and a half: how you eat, which is the vagus nerve at the table

This section exists because the mind-gut driver from Chapter 4 of the ebook shows up at every meal, and almost nobody works on it.

Digestion is a parasympathetic activity. The vagus nerve drives the sequence: it triggers stomach acid and enzyme release in anticipation of food, it coordinates the muscular activity that moves a meal along, and it governs the rate at which the stomach empties. Eat while stressed, rushed, or working, and you are asking the digestive system to run while the nervous system is prioritizing something else.

The nervous system also sets the volume on the symptom. Esophageal hypervigilance is present across reflux presentations regardless of acid burden and predicts symptom severity (PMID 33432708), and in patients undergoing pH monitoring, a psychological stressor increased symptom ratings, with the authors noting that symptom perception **in the absence of increased reflux** may explain poor responses to standard treatment (PMID 16310024). Not every study agrees, and a short laboratory stressor did not increase acid perception in one experiment (PMID 19453516), so the better-supported version is chronic load rather than a single bad afternoon.

What to actually do, in order of how much it matters:

1. **Take five slow breaths before the first bite.** Longer exhale than inhale. This is the cheapest vagal input available and it takes thirty seconds.
2. **Sit down. No screens, no driving, no standing at the counter.** The posture point is not manners, it is abdominal pressure and diaphragm position.
3. **Chew until the texture is gone.** Chewing is the only part of digestion under conscious control, and it is the step that determines how much work the stomach has to do.
4. **Put the fork down between bites.** The single most reliable way to slow a meal without counting anything.
5. **Stop at comfortably satisfied, not full.** Fullness is stomach distension, and distension is the trigger for the sphincter relaxations that produce reflux.
6. **Do not lie down, and do not exercise hard, for the hour after eating.** Walk instead. A gentle walk after a meal supports emptying without raising abdominal pressure the way crunches or heavy lifting does.

Then breathe again afterward. The highest-value time for the diaphragmatic breathing practice in the supplement guide is the half hour after a meal, because that is when reflux events cluster and when the practice was shown to reduce them.

Part three: the eating pattern

A plant-forward Mediterranean pattern

The most interesting study in this area compared a dietary approach against medication. In a retrospective comparison, patients with laryngopharyngeal reflux treated with alkaline water, a plant-based Mediterranean-style diet, and standard reflux precautions were compared with patients treated with proton pump inhibition and standard precautions. The percentage reduction in the reflux symptom index was significantly greater with the dietary approach, and the authors suggest a plant-based diet and alkaline water should be considered in the treatment of this condition (PMID 28880991).

How to read that honestly. It is retrospective rather than randomized, and it is in laryngopharyngeal reflux, the throat-and-voice presentation, rather than classic heartburn. That is a real limitation. It is also one of the few studies where a dietary approach was measured against the drug and did not lose, which is worth knowing.

What the pattern means practically: vegetables and fruit at most meals, olive oil as the main fat, fish and legumes for protein more often than red and processed meat, whole grains, and far less ultra-processed food.

Fiber, and the reason it belongs here

A pattern built on plants brings fiber, and fiber supports regular bowel function. Straining and constipation raise intra-abdominal pressure, which is the same mechanism by which abdominal weight worsens reflux. This is an unglamorous connection that pays off.

Increase fiber gradually. A sudden jump produces bloating, and a distended abdomen is not what you want.

The histamine question, briefly

Some people with reflux also have histamine-related symptoms, and the aged, fermented, and leftover foods that carry the most histamine overlap with the classic reflux trigger list. If flushing, headache, hives, or palpitations travel with your digestive symptoms, the four-week freshness-based trial in the Allergy Diet Guide is the structured way to test that, and DAO support is covered in the supplement guide. If your symptoms are purely heartburn, this is probably not your thread to pull.

Bacterial overgrowth, if bloating dominates

A case-control study of 50 people with GERD confirmed by manometry and acid monitoring, against 53 symptom-matched controls, found small intestinal bacterial overgrowth positively correlated with GERD and higher mean expiratory hydrogen in the GERD group (PMID 41131458). That is an association in 103 people, not a proven mechanism.

Practically: if bloating and distension are as prominent as the burning, it is worth raising breath testing with a clinician rather than reaching for a low FODMAP diet on your own. Low FODMAP is a demanding, temporary, dietitian-supported protocol, and it is not a reflux diet.

A week that supports the plan

The rhythm is the point: three moderate meals, the last one finished three hours before bed.

Breakfast options. Oats with berries and ground flax. Eggs with sauteed greens. Plain yogurt with fruit if dairy is not on your personal list.

Lunch options. Lentil or vegetable soup with olive oil. Chicken or fish over rice or quinoa with vegetables. A large salad with beans, olive oil, and a protein.

Dinner options, finished early. Baked fish with potatoes and greens. Chicken thighs with roasted vegetables. Turkey and vegetable stir fry over rice.

The rules that matter more than the menu. Moderate portions. Last bite three hours before lying down. Left side, head of bed raised. Alcohol out during any trial period. Water through the day rather than large volumes with meals.

What I deliberately left out, and why

- **The universal trigger list as a prescription.** No published evidence of efficacy for dietary measures in the evidence review (PMID 16682569). Use it as candidates to test, not rules to obey.
- **Baking soda in water as a routine remedy.** It relieves symptoms briefly and it is a large sodium load, it can cause rebound, and in rare cases stomach rupture has been described with large amounts on a full stomach. Alginate products are covered in the supplement guide and are a better answer.
- **Apple cider vinegar as a general reflux treatment.** Widely recommended on the low-acid theory. I could not locate randomized trial evidence for reflux, and if there is an ulcer or erosive esophagitis it can hurt. It is not on this list on the strength of theory alone.
- **Long elimination diets.** Twelve removed foods is a method failure, not a discovery.
- **Alkaline water as a stand-alone answer.** It appeared as one component of a dietary package in a retrospective study (PMID 28880991). That is not the same as the water doing the work.

Telling you what to skip is part of what you paid for.

When to seek professional care

Emergency: chest pain with sweating, nausea, breathlessness, or pain radiating to the jaw or arm. Vomiting blood or black tarry stools. Inability to swallow saliva.

Soon:

- Difficulty swallowing or food sticking, ever
- Unintentional weight loss, persistent vomiting, or anemia
- New symptoms at age 60 or over
- Symptoms requiring daily treatment for more than a few weeks
- Many years of untreated symptoms, which is a Barrett’s screening conversation
- Reflux with cough, hoarseness, or worsening asthma

Working with Root Health

This guide is written to be safe and useful for a reader I have never met. A personalized plan starts from your diary, your test results, and your medication list, and sequences changes rather than stacking them.

> ***Book a free Root Discovery Session: rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=reflux-diet***

Consultations are available to people located in North Carolina, where Dr. Seay is licensed.

If you are outside North Carolina, the tracker is what turns a food experiment into an answer:

> ***Free symptom tracker: rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=reflux-diet***

Read the companion documents alongside this one. **Understanding Your Reflux Labs** covers H. pylori and the conditions that imitate reflux, and **The Reflux Supplement Guide** covers what has been studied and how to come off acid suppression safely.

References

All references were verified against live PubMed records on 2026-08-08, including a publication type and corrections check for retractions and errata. Verification detail is documented in `Reflux-Kit-Citation-Ledger.md`.

1. Kaltenbach T, Crockett S, Gerson LB. Are lifestyle measures effective in patients with gastroesophageal reflux disease? An evidence-based approach. *Arch Intern Med.* 2006;166(9):965-971. PMID 16682569

2. Ness-Jensen E, Hveem K, El-Serag H, et al. Lifestyle Intervention in Gastroesophageal Reflux Disease. *Clin Gastroenterol Hepatol.* 2016;14(2):175-182. PMID 25956834

3. Ness-Jensen E, Lindam A, Lagergren J, et al. Tobacco smoking cessation and improved gastroesophageal reflux: a prospective population-based cohort study: the HUNT study. *Am J Gastroenterol.* 2014;109(2):171-177. PMID 24322837

4. Zalvan CH, Hu S, Greenberg B, et al. A Comparison of Alkaline Water and Mediterranean Diet vs Proton Pump Inhibition for Treatment of Laryngopharyngeal Reflux. *JAMA Otolaryngol Head Neck Surg.* 2017;143(10):1023-1029. PMID 28880991

5. Hu Z, Shan B, Wang Y, et al. Association of gastroesophageal reflux disease with small intestinal bacterial overgrowth: a case-control study. *BMC Gastroenterol.* 2025;25(1):753. PMID 41131458

6. Guadagnoli L, Yadlapati R, Taft T, et al. Esophageal hypervigilance is prevalent across gastroesophageal reflux disease presentations. *Neurogastroenterol Motil.* 2021;33(8):e14081. PMID 33432708

7. Wright CE, Ebrecht M, Mitchell R, et al. The effect of psychological stress on symptom severity and perception in patients with gastro-oesophageal reflux. *J Psychosom Res.* 2005;59(6):415-424. PMID 16310024

8. Hemmink GJ, Bredenoord AJ, Weusten BL, et al. Does acute psychological stress increase perception of oesophageal acid? *Neurogastroenterol Motil.* 2009;21(10):1055-e86. PMID 19453516

9. Tan X, Wang S, Wu F, et al. Bidirectional correlation between gastroesophageal reflux disease and sleep problems: a systematic review and meta-analysis. *PeerJ.* 2024;12:e17202. PMID 38646475

Dr. Adam Seay, DC · Seay Wellness / Root Health, Garner, NC
rootcauseunlocked.com · (919) 662-0520

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Call 911 for chest pain or pressure with sweating, nausea, breathlessness, or pain into the jaw or arm.

See a doctor before treating yourself if you have difficulty swallowing, food sticking, unintentional weight loss, vomiting blood, black tarry stools, anemia, or new symptoms after age 60.

Do not stop a proton pump inhibitor abruptly on the strength of this document. There is a section below on how deprescribing is actually done, and the short version is that stopping suddenly is the version that tends to fail. If you take a PPI for Barrett's esophagus, a bleeding ulcer, or gastroprotection alongside NSAIDs, aspirin, or a blood thinner, that prescription is doing a specific job and stopping it is a decision for the prescriber, not for a guide.

Supplements in the United States are regulated as food, not as drugs. Quality varies. Look for independent third-party testing.

Why nutrient evidence looks thinner than it is

A large outcome trial costs tens to hundreds of millions of dollars, and that money only exists when a patent can pay it back. You cannot patent magnesium, an herb, or a food. So the evidence hierarchy this guide grades against is not neutral: it systematically over-documents patentable drugs and under-documents nearly everything else. That is measurable, not rhetorical. A Cochrane methodology review found that manufacturer sponsorship of drug and device studies produces more favourable results and conclusions than independent funding, through a bias that standard quality checks do not catch (PMID 28207928).

Two honest consequences follow, and they pull in opposite directions.

First, "no evidence" has three different meanings, and this guide tries to always tell you which one applies. *Tested and failed*: a real trial ran and the answer was no. *Never properly tested*: nobody could afford the trial, so absence of evidence is not evidence of absence. *Tested small and won*: real but modest trials, usually measuring markers rather than outcomes, because markers are what a small budget can afford.

Second, the asymmetry earns nutrients humility, not automatic credit. When isolated nutrients have received large publicly funded trials, the results have been mixed: some genuine wins, some nulls, and some harms. Small nutrient trials also carry their own publication bias toward positive results, just as drug trials tilt toward their sponsor. So this guide applies one rule in both directions: name the funding, name the evidence tier, and let you see both.

Where supplements actually fit in reflux

Reflux has five drivers, and this kit is organized around them: **the barrier** (the valve and the diaphragm), **the signal** (vagal tone, stress, sleep), **the terrain** (the microbial environment), **the load** (meals, weight, timing), and **the capacity** (whether the stomach is doing its job).

Here is the honest placement of the products in this guide: **most of them act on the consequence, not on a driver.** Acid is what makes reflux hurt. It is almost never what makes reflux happen. Soothing and neutralizing has real value when you are in pain, and it does not repair a barrier or retrain a nervous system.

That is not a counsel of despair, because two of the five drivers respond to things you can do yourself, for free, starting today. They come first in this guide for that reason, ahead of anything you can buy.

The two acid-suppressing realities to keep in view. Up to half of people remain symptomatic despite adequate acid suppression, which is itself the argument that reflux is multifactorial rather than acid-centric (PMID 42198379). And the interventions with the best evidence in the whole kit, weight loss and head-of-bed elevation, are dietary and mechanical rather than pharmacological (PMID 16682569).

So the order is: work the barrier, work the signal, fix the terrain if it is disturbed, reduce the load, and use the products below to make all of that survivable while it takes effect.

How to read this guide

The tiers. Tier 1 has randomized human evidence in reflux specifically. Tier 2 has evidence in adjacent conditions such as functional dyspepsia, or in reflux with real limitations. Tier 3 needs a prescriber conversation first.

Track two numbers daily: symptom severity out of ten, and how many doses of any rescue product you took. The second number moves before the first.

One change at a time, and give most of these four weeks.

Foundation 1: retrain the diaphragm, because it is half of your sphincter

This is first because it is free, because it acts on the mechanism rather than the symptom, and because most people have never been told it.

The physiology. The anti-reflux barrier has three parts: the lower esophageal sphincter, the crural diaphragm wrapped around it, and the flap valve at the entry angle (PMID 40412989). The crural diaphragm functions as an external sphincter, raising sphincter pressure with every breath in and during any rise in abdominal pressure (PMID 15316404). Most reflux happens through transient sphincter relaxations rather than a permanently open valve, and crural support is a real part of what holds the line (PMID 11003803).

The trial. In patients with pH-proven upright reflux, diaphragmatic breathing reduced the number of postprandial reflux events by increasing the difference between lower esophageal sphincter and gastric pressure. Esophageal acid exposure in the two hours after a standardized meal fell from 11.8% to 5.2% (PMID 33009052). A 2026 systematic review and meta-analysis found a significant reduction in symptom scores favoring diaphragmatic breathing, while stating plainly that heterogeneity was substantial and the evidence is not yet sufficient for definitive recommendations (PMID 42123139).

A structured inspiratory muscle training trial aimed at this exact mechanism is registered and published as a protocol, with results not yet reported (PMID 41874149). Promising, pending, and I would rather say so.

The practice. Sitting or lying, one hand on the chest and one on the abdomen. Breathe so the lower hand rises and the upper hand stays still. Exhale longer than you inhale. Five to ten minutes, twice daily, and **the highest-value window is the half hour after a meal**, because that is when reflux events cluster.

The structural half, which is my own field. If the diaphragm is half the sphincter, then thoracic mobility, rib position, posture, and how you manage abdominal pressure under load are all inputs to the valve. Bending, lifting badly, straining, tight waistbands, and a slumped seated posture all raise the pressure your barrier has to hold against. That is why the postural chapter of the ebook is not filler, and why a structural assessment belongs in a reflux workup.

Foundation 2: work the nervous system, which sets the volume

Why it belongs in a supplement guide. Because it outperformed most of the products below in the one trial that measured an objective endpoint.

In a controlled study of 20 subjects with documented reflux disease, relaxation training during a stressful task produced significantly lower heart rate and anxiety ratings than an attention-placebo control, and significantly lower reflux symptom ratings **and lower total esophageal acid exposure** (PMID 8020690). A relaxation intervention moved measured acid exposure, not just how people felt about it.

Why it works. Esophageal hypervigilance is present across reflux presentations regardless of acid burden and independently predicts symptom severity (PMID 33432708). Stress increases symptom ratings, and the authors of one pH-monitored study specifically note that symptom perception in the absence of increased reflux may explain poor responses to standard treatment (PMID 16310024). The honest counterweight: a short experimental stressor did not increase acid perception in another study (PMID 19453516), so chronic load has better support than acute stress does.

Sleep is the other half of this driver, and it runs in both directions with reflux: poor sleep quality, sleep disturbance, and short sleep duration are all associated with reflux, bidirectionally (PMID 38646475). If you snore or wake unrefreshed, ask about sleep apnea, which is linked to reflux through airway irritation and vagal pathways (PMID 40045407). Treating apnea can improve reflux, and nothing in this guide substitutes for that.

What to actually do. A daily relaxation practice you will repeat, which matters more than which one. The breathing practice above does double duty here, since it is both a mechanical input to the diaphragm and a vagal one. Protect sleep as a treatment rather than a luxury. And read the eating mechanics section of the diet guide, which applies the same driver at the table.

Then the products: Tier 1, studied in reflux

Alginate

What it is. A seaweed-derived compound that forms a physical raft floating on top of the stomach contents. It is the one item in this guide that acts mechanically rather than chemically, which is why I put it first.

What the research shows. A systematic review and meta-analysis of 14 studies covering 2,095 subjects found alginate-based therapies increased the odds of resolution of reflux symptoms compared with placebo or antacids, with an odds ratio of 4.42 (95% CI 2.45 to 7.97), with moderate heterogeneity. Compared with PPIs or H2 blockers, alginates appeared less effective, though that pooled estimate was not statistically significant (PMID 28375448).

A second systematic review and meta-analysis of randomized trials found alginates may have greater efficacy than placebo or antacids, described the evidence on alginate versus PPI as questionable and suggesting no difference, and found the risk of adverse events no greater than placebo or PPIs (PMID 33275256). A broader meta-analysis of over-the-counter reflux therapies found alginate and antacid combinations superior to placebo in symptomatic improvement, with an absolute benefit increase of 26% (PMID 17229239).

How to read that. Better than placebo and better than plain antacids, with a clean safety profile, and probably not equal to a PPI. That is a genuinely useful place in the toolkit, particularly for breakthrough symptoms and for people trying to reduce acid suppression.

Typical use. After meals and at bedtime, which is when the raft is most useful. It works best taken after eating rather than before.

Cautions. Most products carry a meaningful sodium load, which matters in heart failure or on a sodium-restricted diet. It can bind other medications, so separate it by two hours.

Melatonin

What it is. Better known for sleep, and it also has effects on gastric acid secretion and on the lower esophageal sphincter.

What the research shows. In a randomized double-blind trial, 78 patients with reflux received omeprazole 20 mg daily plus either sublingual melatonin 3 mg or placebo for four weeks. Symptom scores declined significantly more in the melatonin group, quality of life scores were significantly higher, and adverse events were similar between groups with none serious (PMID 37768310).

An earlier comparative study tested a supplement containing melatonin, l-tryptophan, vitamin B6, folic acid, vitamin B12, methionine, and betaine against omeprazole 20 mg in 351 patients. All 176 patients in the supplement group reported complete regression of symptoms after 40 days, compared with 65.7% in the omeprazole group (PMID 16948779).

How to read that second study honestly. A 100% response rate is an extraordinary claim, it was a multi-ingredient formula so no single component can be credited, and it has not been replicated at that magnitude. I include it because it exists and because the melatonin literature in reflux rests partly on it, not because I expect you to get that result.

Typical studied range. 3 mg sublingually at night, as an addition to standard treatment (PMID 37768310).

Cautions. Next-morning grogginess usually means too high a dose. It adds to other sedating medication. Not established in pregnancy or nursing.

Tier 2: evidence in adjacent conditions, or with real limits

Iberogast and similar multi-herb formulas

What it is. A standardized nine-herb liquid preparation used for decades in functional dyspepsia, built around bitter candytuft with chamomile, peppermint, caraway, licorice, lemon balm, celandine, angelica, and milk thistle.

What the research shows. A narrative review describes it as an evidence-based phytomedicine with a favorable safety profile, addressing motility, hypersensitivity, inflammation, and barrier function, and specifically names overlapping dyspepsia and reflux phenotypes as an area for future research (PMID 41319263). In a randomized, double-blinded, placebo-controlled study of 32 patients with functional dyspepsia and bloating, two weeks of the STW 5-II formulation produced significantly lower symptom perception during a gastric gas challenge than placebo, with no difference in abdominal distension (PMID 40684457).

How to read it. Good evidence in functional dyspepsia, which overlaps with reflux without being the same condition. Reasonable if bloating and fullness after meals are as prominent as burning.

Cautions. It contains peppermint, which is traditionally avoided in reflux on sphincter-relaxation grounds, so this is one to test rather than assume. Regulatory authorities have flagged rare liver injury with this class of preparation, so stop and call for jaundice, dark urine, or right-sided abdominal pain. Contains licorice and alcohol in the liquid form.

Curcumin

What it is. The active fraction of turmeric. Poorly absorbed on its own, which is why formulation matters more here than for almost anything else in this guide.

What the research shows. A randomised, double blind controlled trial in Thailand assigned 206 patients with functional dyspepsia to curcumin alone, omeprazole alone, or both, for 28 days, with follow-up to day 56. Symptoms were scored with the Severity of Dyspepsia Assessment. All three groups improved significantly on the pain, non-pain, and satisfaction categories at day 28, and improved further by day 56. There were **no significant differences between the three groups**, no synergistic effect from combining them, and no serious adverse events (PMID 37696679).

Headline version: curcumin performed comparably to omeprazole. That is a striking result for a botanical, and it is the reason this entry exists.

Now the caveat that decides how much weight to give it, and it is the one every summary of this trial leaves out. There was **no placebo arm.** All three groups got an active treatment, and all three improved. Functional dyspepsia has a large placebo response and a naturally fluctuating course, so a three-arm trial in which everyone improves and nobody differs cannot separate “both treatments work equally well” from “both treatments are tracking the same placebo and regression-to-the-mean effect.” Those two readings are very different and this design does not distinguish them.

Two more things worth knowing. 206 patients enrolled and 151 completed, so roughly a quarter left the study, which is high. And the paper carries a published correction from 2025 (PMID 38866470). A correction is not a retraction and the record is in good standing, but you should hear it from me.

Why it is in Tier 2 rather than Tier 1. The trial was in **functional dyspepsia**, not reflux. Those conditions overlap and are not the same: functional dyspepsia is epigastric pain and fullness, reflux is heartburn and regurgitation. Same reason Iberogast sits in this tier.

Typical studied range. 250 mg four times daily, taken as two capsules per dose, giving 2 g daily for 28 days (PMID 37696679). That is a higher and more frequent dose than most people take, and it is what was tested.

Cautions. Curcumin increases bleeding risk with anticoagulants and antiplatelet drugs, and should be stopped before surgery. It can cause reflux and stomach upset in some people, which matters in this guide specifically. It interacts with several drugs through the same liver enzymes, and it is not established in pregnancy. Plain turmeric powder is not the same as the standardised extract used in the trial.

Zinc L-carnosine

What it is. A zinc compound with an unusual property: it binds preferentially to damaged mucosa and releases zinc locally.

What the research shows. A review of its role in gastrointestinal mucosal disease describes direct mucosal cytoprotective and anti-inflammatory action through antioxidative effects, cytokine modulation, and membrane stabilization, notes that it is not an antacid or acid-suppressing agent, and describes promotion of repair of mucosal injury in humans across several settings (PMID 35659631).

How to read it. Mucosal repair rather than symptom suppression, and the human literature is broader in gastric injury than in reflux specifically. Sensible where the goal is healing an irritated lining rather than blocking acid.

Cautions. Zinc long-term can deplete copper. Do not run high-dose zinc indefinitely without monitoring.

Ginger

What the research shows. In a randomized double-blind crossover study of 11 patients with functional dyspepsia, ginger accelerated gastric emptying, with median half-emptying time of 12.3 minutes versus 16.1 minutes on placebo, and showed a trend toward more antral contractions, while **gastrointestinal symptoms did not differ** (PMID 21218090). A pilot randomized crossover study in 11 healthy volunteers found a standardized ginger and artichoke combination significantly promoted gastric emptying compared with placebo (PMID 26813467).

How to read it. It moves the stomach along, which is a sensible mechanism if slow emptying is part of your picture. It did not change symptoms in the dyspepsia trial, and both studies had 11 participants. Cheap and low risk, with modest expectations.

Cautions. Can increase bleeding risk at higher doses with anticoagulants. Commonly causes heartburn in some people, which is worth noting in a reflux guide.

D-limonene

Widely sold for heartburn. The most-cited source is a review describing it as generally recognized as safe as a flavoring agent, with low toxicity, and noting that because of its gastric acid neutralizing effect and support of normal peristalsis it has been used for relief of heartburn and reflux (PMID 18072821).

Be clear about what that is. It is a safety and applications review describing use, not a randomized trial demonstrating benefit. I could not locate controlled trial evidence in reflux. Low risk, plausible, unproven. If you try it, know which of those three words is doing the work.

Probiotics, mainly as an adjunct to H. pylori treatment

This is where the probiotic evidence in this category is strongest, and it is about tolerating the antibiotics rather than treating reflux.

A meta-analysis of Saccharomyces boulardii as an adjuvant to standard eradication therapy found it significantly improved eradication rates and reduced total adverse effects (RR 0.49), diarrhea (RR 0.36), abdominal distension (RR 0.49), constipation (RR 0.38), and nausea (RR 0.50), while not reducing abdominal pain, vomiting, or taste disturbance (PMID 40012609). A meta-analysis of eight randomized trials in 1,087 patients found Lactobacillus reuteri supplementation tended to increase eradication rates, reduce antibiotic-related side effects, and alleviate gastrointestinal symptoms during treatment (PMID 38846173).

Practical use. If you are going through eradication therapy, this is worth raising with your prescriber. It is not a reflux treatment on its own.

Tier 3: ask your prescriber first

Betaine HCl and the low stomach acid question

The lab guide covers the evidence in full. The summary: the strongest published case for replacing lost stomach acid is made specifically for **autoimmune gastritis**, a defined condition with documented destruction of the acid-producing cells, where the argument is that B12 and iron are routinely replaced while the missing acid is not, and where re-acidification may improve gastrointestinal symptoms (PMID 38474790).

Where that leaves betaine HCl. It has a coherent rationale in someone with demonstrated hypochlorhydria. It does not have trial evidence supporting its use across reflux generally, and I am not going to pretend otherwise in a document you paid for.

Why it is in Tier 3 rather than Tier 1. Because if you have an ulcer, erosive esophagitis, or active gastritis, adding acid can hurt you, and those are exactly the conditions that produce the symptoms leading people to try it. **Do not self-start betaine HCl if you take NSAIDs, aspirin, or steroids, if you have a known ulcer, or if you have alarm features.** If low acid is genuinely suspected, the honest sequence is to demonstrate it first with the workup in the lab guide, then decide with a clinician.

Deglycyrrhizinated licorice

Traditionally used for mucosal soothing. I searched for randomized evidence in reflux and functional dyspepsia and did not find trials that would justify a strong claim, so it sits here as low risk and unproven rather than as a recommendation. The deglycyrrhizinated form exists specifically to remove the compound that raises blood pressure and lowers potassium, so if you use licorice, use that form. Standard licorice is not interchangeable and matters with hypertension, heart disease, and diuretics.

Coming off a PPI, which is what most readers actually came here for

This deserves its own section because it is where the most harm gets done by well-meaning advice, in both directions.

The terrain argument, stated fairly. A narrative review from a microbial ecology perspective notes that up to 50% of people remain symptomatic despite adequate acid suppression, describes reflux as multifactorial rather than acid-centric, and examines how long-term acid suppression alters the microbial ecology of the upper gut, while being explicit that causality is not fully established (PMID 42198379). That is a reasonable argument for not staying on acid suppression by default. It is not an argument for stopping one abruptly, and the distinction matters.

First, the honest evidence on rebound. In a randomized, double-blind, placebo-controlled trial, 120 **healthy volunteers with no reflux** received either 12 weeks of placebo or 8 weeks of esomeprazole 40 mg followed by 4 weeks of placebo. In weeks 9 to 12, significantly more of the PPI group reported dyspepsia, heartburn, or acid regurgitation than the placebo group, around 21% to 22% versus 2% to 7% (PMID 19362552).

Read that carefully: people who never had reflux developed reflux symptoms after stopping a PPI. That is why stopping feels like proof you still need it when it may be proof of nothing.

And the counterweight, because it exists. A systematic review of rebound acid hypersecretion found the studies heterogeneous in design and outcome, with some evidence from uncontrolled trials of increased acid-secreting capacity in H. pylori-negative subjects after 8 weeks, and concluded there is **no strong evidence for a clinically relevant increased acid production** after withdrawal (PMID 17229219).

So the symptoms on withdrawal are well demonstrated; whether the underlying acid physiology is truly rebounding is less settled. Either way, the practical implication is the same.

How deprescribing is actually done. A 2025 evidence-based review of PPI indications, harms, and deprescribing describes the standard approaches: tapering the dose, switching to on-demand or intermittent use, and switching to an H2 receptor antagonist, particularly in people with non-erosive reflux disease or functional dyspepsia, with formal tools available to identify candidates for tapering (PMID 41010960).

A structure to bring to your prescriber, not to run alone:

1. **Confirm you are a candidate.** Not if you have Barrett's, a history of bleeding ulcer, or need gastroprotection with NSAIDs or antiplatelet therapy.
2. **Get the foundations in place first**, for at least two to four weeks: the head of the bed raised, three hours between eating and lying down, weight addressed if relevant, alcohol reduced, and the breathing practice started.
3. **Taper rather than stop.** Halve the dose, or move to alternate days, over several weeks.
4. **Have a bridge ready.** This is what alginate is for, and an H2 blocker is the other option your prescriber may use.
5. **Expect two to four weeks of worse symptoms**, and know in advance that this is the documented withdrawal pattern rather than proof of failure (PMID 19362552).
6. **Reassess honestly.** If symptoms genuinely return and stay, the medication was doing a job. That is a legitimate outcome, not a personal failure.

What I deliberately left out, and why

- **Baking soda as a routine remedy.** Fast relief, large sodium load, rebound, and rare reports of serious harm with large amounts. Alginate does the same job better.
- **Apple cider vinegar.** Widely recommended on the low-acid theory. No randomized reflux evidence located, and a real potential to hurt if there is an ulcer or erosive disease.
- **Aloe vera juice.** Some early work in reflux exists, and the products vary enormously and some have a laxative effect from the latex fraction. Not enough to recommend by name.
- **Slippery elm and marshmallow root.** Soothing, harmless, traditional. I could not locate randomized reflux trials, so they are named here honestly rather than sold to you.
- **Digestive enzyme blends for everyone.** Reasonable in demonstrated insufficiency, which is a different diagnosis. Not a general reflux answer.
- **Stopping your PPI cold because a supplement guide said acid is good.** The most common harm in this category.

Telling you what to skip is part of what you paid for.

Realistic expectations

What is realistic. Fewer breakthrough episodes, fewer rescue doses, sleeping through the night, and for some people getting off or down on daily medication with a plan and a prescriber.

What is not realistic. A supplement fixing the mechanics. If the sphincter and the anatomy are the problem, chemistry only manages it.

Timeline. Four weeks for most of these. The mechanical changes, meal timing and bed elevation, often work within days, which is another reason to do them first.

Cost sense. If a product has done nothing after four weeks, stop paying for it. Alginate and a wedge under the mattress cost less than most of the shelf.

When to seek professional care

Emergency: chest pain with sweating, nausea, breathlessness, or pain to the jaw or arm. Vomiting blood or black tarry stools. Inability to swallow saliva.

Soon:

- Difficulty swallowing or food sticking, ever
- Unintentional weight loss, persistent vomiting, or anemia
- New symptoms at 60 or over
- Daily symptoms for more than a few weeks despite treatment
- Many years of untreated reflux, which is a Barrett's screening conversation
- Any plan to come off long-term acid suppression

Stop and call your prescriber for: yellowing of the skin or eyes, dark urine, or right-sided abdominal pain while taking a multi-herb preparation. Also for new black stools, vomiting blood, or worsening pain after starting anything acidic.

Where to find quality versions of these

Nothing here names a brand, on purpose. The evidence supports ingredients and forms, not labels.

Look for third-party testing. NSF, USP Verified, or an equivalent independent seal.

Check the form against the studied one. Sublingual melatonin at 3 mg. Alginate rather than a plain antacid, and check the sodium content. Zinc L-carnosine rather than generic zinc, since the compound is the point.

Read the whole label. Multi-herb digestive formulas frequently include peppermint, which is one of the things you may be trying to avoid.

If you would rather not evaluate all of that yourself, Root Health keeps a curated dispensary of professional-grade products, screened for the standards above and organized by the categories in this guide.

> ***Root Health dispensary: us.fullscript.com/welcome/drseay***
>
> *Disclosure: Root Health earns a portion of dispensary purchases. That revenue supports the practice, and it is also why this guide was written to be complete and usable on its own. Nothing here was included, excluded, ranked, or dosed based on what it earns. You can buy any of it anywhere, and the highest-value items in this guide, raising the head of your bed, eating earlier, and the breathing practice, cost nothing at all.*

Working with Root Health

Everything here is general education, written to be safe for a reader I have never examined. A personalized plan starts from your history, your test results, and your full medication list.

> ***Book a free Root Discovery Session: rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=reflux-supplement***

Consultations are available to people located in North Carolina, where Dr. Seay is licensed.

If you are outside North Carolina, track first:

> ***Free symptom tracker: rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=reflux-supplement***

Read the companion documents alongside this one. **Understanding Your Reflux Labs** covers H. pylori, the imitators, and the workup for genuinely low stomach acid, and **The Reflux Diet Guide** covers the two measures with the best evidence of anything in this kit.

References

All references were verified against live PubMed records on 2026-08-08, including a publication type and corrections check for retractions and errata. Verification detail is documented in `Reflux-Kit-Citation-Ledger.md`.

1. Kaltenbach T, Crockett S, Gerson LB. Are lifestyle measures effective in patients with gastroesophageal reflux disease? An evidence-based approach. *Arch Intern Med.* 2006;166(9):965-971. PMID 16682569
2. Leiman DA, Riff BP, Morgan S, et al. Alginate therapy is effective treatment for GERD symptoms: a systematic review and meta-analysis. *Dis Esophagus.* 2017;30(5):1-9. PMID 28375448
3. Zhao CX, Wang JW, Gong M. Efficacy and safety of alginate formulations in patients with gastroesophageal reflux disease: a systematic review and meta-analysis of randomized controlled trials. *Eur Rev Med Pharmacol Sci.* 2020;24(22):11845-11857. PMID 33275256
4. Tran T, Lowry AM, El-Serag HB. Meta-analysis: the efficacy of over-the-counter gastro-oesophageal reflux disease therapies. *Aliment Pharmacol Ther.* 2007;25(2):143-153. PMID 17229239
5. Malekpour H, Noori A, Abdi S, et al. Is the Addition of Sublingual Melatonin to Omeprazole Superior to Omeprazole Alone in the Management of Gastroesophageal Reflux Disease Symptoms: A Clinical Trial. *Turk J Gastroenterol.* 2023;34(12):1206-1211. PMID 37768310
6. Pereira Rde S. Regression of gastroesophageal reflux disease symptoms using dietary supplementation with melatonin, vitamins and aminoacids: comparison with omeprazole. *J Pineal Res.* 2006;41(3):195-200. PMID 16948779
7. Halland M, Bharucha AE, Crowell MD, et al. Effects of Diaphragmatic Breathing on the Pathophysiology and Treatment of Upright Gastroesophageal Reflux: A Randomized Controlled Trial. *Am J Gastroenterol.* 2021;116(1):86-94. PMID 33009052
8. Abureesh O, Qaqish F, Abu-Shaban M, et al. Efficacy and Safety of Diaphragmatic Breathing Exercises for Gastroesophageal Reflux Disease: A Systematic Review and Meta-Analysis. *J Clin Med.* 2026;15(9). PMID 42123139
9. Lambiase C, Usai Satta P, Bassotti G, et al. Iberogast in functional dyspepsia: yesterday, today, and tomorrow, a narrative review of a multitarget phytomedicine. *Expert Opin Pharmacother.* 2025;26(17):1867-1876. PMID 41319263
10. Aguilar A, Alcala-Gonzalez L, Barber C, et al. Effect of STW 5-II (Iberogast-N) on Tolerance to Gastric Gas in Patients With Functional Dyspepsia. The IBO-2 Study. *Neurogastroenterol Motil.* 2025;37(11):e70123. PMID 40684457
11. Efthymakis K, Neri M. The role of Zinc L-Carnosine in the prevention and treatment of gastrointestinal mucosal disease in humans: a review. *Clin Res Hepatol Gastroenterol.* 2022;46(7):101954. PMID 35659631
12. Hu ML, Rayner CK, Wu KL, et al. Effect of ginger on gastric motility and symptoms of functional dyspepsia. *World J Gastroenterol.* 2011;17(1):105-110. PMID 21218090
13. Lazzini S, Polinelli W, Riva A, et al. The effect of ginger (Zingiber officinalis) and artichoke (Cynara cardunculus) extract supplementation on gastric motility: a pilot randomized study in healthy volunteers. *Eur Rev Med Pharmacol Sci.* 2016;20(1):146-149. PMID 26813467
14. Sun J. D-Limonene: safety and clinical applications. *Altern Med Rev.* 2007;12(3):259-264. PMID 18072821
15. Li M, Xie Y. Efficacy and safety of Saccharomyces boulardii as an adjuvant therapy for the eradication of Helicobacter pylori: a meta-analysis. *Front Cell Infect Microbiol.* 2025;15:1441185. PMID 40012609
16. Li M, Wang X, Dong X, et al. Lactobacillus reuteri compared with placebo as an adjuvant in Helicobacter pylori eradication therapy: a meta-analysis of randomized controlled trials. *Ther Adv Gastroenterol.* 2024;17:17562848241258021. PMID 38846173

17. Taylor L, McCaddon A, Wolffenbuttel BHR. Creating a Framework for Treating Autoimmune Gastritis, The Case for Replacing Lost Acid. *Nutrients.* 2024;16(5):662. PMID 38474790

18. Reimer C, Sondergaard B, Hilsted L, et al. Proton-pump inhibitor therapy induces acid-related symptoms in healthy volunteers after withdrawal of therapy. *Gastroenterology.* 2009;137(1):80-87. PMID 19362552

19. Hunfeld NG, Geus WP, Kuipers EJ. Systematic review: Rebound acid hypersecretion after therapy with proton pump inhibitors. *Aliment Pharmacol Ther.* 2007;25(1):39-46. PMID 17229219

20. Andrawes M, Andrawes W, Das A, et al. Proton Pump Inhibitors, An Evidence-Based Review of Indications, Efficacy, Harms, and Deprescribing. *Medicina (Kaunas).* 2025;61(9). PMID 41010960

21. Kahrilas PJ, Nguyen NT. The Esophagogastric Junction Reconsidered. *Gastrointest Endosc Clin N Am.* 2025;35(3):541-553. PMID 40412989

22. van Herwaarden MA, Samsom M, Smout AJ. The role of hiatus hernia in gastro-oesophageal reflux disease. *Eur J Gastroenterol Hepatol.* 2004;16(9):831-835. PMID 15316404

23. Holloway RH. The anti-reflux barrier and mechanisms of gastro-oesophageal reflux. *Baillieres Best Pract Res Clin Gastroenterol.* 2000;14(5):681-699. PMID 11003803

24. Syropoulos S, Moutzouri M, Grammatopoulou E, et al. Impact of Inspiratory Muscle Training in Individuals with Gastroesophageal Reflux Disease: A Randomized Controlled Trial Protocol. *Methods Protoc.* 2026;9(2). PMID 41874149

25. McDonald-Haile J, Bradley LA, Bailey MA, et al. Relaxation training reduces symptom reports and acid exposure in patients with gastroesophageal reflux disease. *Gastroenterology.* 1994;107(1):61-69. PMID 8020690

26. Guadagnoli L, Yadlapati R, Taft T, et al. Esophageal hypervigilance is prevalent across gastroesophageal reflux disease presentations. *Neurogastroenterol Motil.* 2021;33(8):e14081. PMID 33432708

27. Wright CE, Ebrecht M, Mitchell R, et al. The effect of psychological stress on symptom severity and perception in patients with gastro-oesophageal reflux. *J Psychosom Res.* 2005;59(6):415-424. PMID 16310024

28. Hemmink GJ, Bredenoord AJ, Weusten BL, et al. Does acute psychological stress increase perception of oesophageal acid? *Neurogastroenterol Motil.* 2009;21(10):1055-e86. PMID 19453516

29. Tan X, Wang S, Wu F, et al. Bidirectional correlation between gastroesophageal reflux disease and sleep problems: a systematic review and meta-analysis. *PeerJ.* 2024;12:e17202. PMID 38646475

30. Wang J, Wang P, Lv J, et al. Exploring the silent connection: unveiling the intricate relationship between gastroesophageal reflux disease and sleep apnea syndrome. *Hum Genomics.* 2025;19(1):23. PMID 40045407

31. Zanoni A, Facchin S, Mari V, et al. Rethinking Long-Term PPI Therapy in GERD: A Narrative Review from a Microbial Ecology Perspective Beyond Acid Suppression. *Pharmaceuticals (Basel).* 2026;19(5). PMID 42198379

32. Kongkam P, Khongkha W, Lopimpisuth C, et al. Curcumin and proton pump inhibitors for functional dyspepsia: a randomised, double blind controlled trial. *BMJ Evid Based Med.* 2023;28(6):399-406. PMID 37696679

33. Lundh A, Lexchin J, Mintzes B, et al. Industry sponsorship and research outcome. *Cochrane Database Syst Rev.* 2017;2(2):MR000033. PMID 28207928

Dr. Adam Seay, DC · Seay Wellness / Root Health, Garner, NC
rootcauseunlocked.com · (919) 662-0520

Educational content only. Not medical advice. Not a diagnosis. Not a substitute for your clinician. Never stop or change a prescribed medication without your prescriber.